

DSAA 1001 Week 2 Validation Set

Probability, Distributions, and Transformations

Instructions

- Purpose: Self-assessment and practice before test
- Time: 30 minutes | Total: 120 pts
- Open book, AI tools allowed
- References: Learning Sheet 1, Sections 1-8

Part A: Multiple Choice [40 points]

Question 1. (5 pts) Which object is an event in a probability space?

- a) a single numerical value of a random variable
- b) a subset of the sample space
- c) the mean of a distribution
- d) a likelihood function viewed without a model

Answer: _____

Question 2. (5 pts) Let $\Omega = \{r, s, t\}$ with probabilities 0.25, 0.50, 0.25. If $X(r) = 2$, $X(s) = 2$, and $X(t) = 5$, what is $P(X = 2)$?

- a) 0.25
- b) 0.50
- c) 0.75
- d) 1

Answer: _____

Question 3. (5 pts) A binomial PMF with parameters (n, p) relies essentially on which assumption?

- a) The trials have independent outcomes with a common success probability
- b) The number of trials is itself Gaussian
- c) The success probability changes after every trial
- d) All possible counts are equally likely

Answer: _____

Question 4. (5 pts) A screening test has a rare condition as its cause. Which statement best describes the base-rate issue in a posterior calculation?

- a) A large likelihood always makes the posterior large
- b) The prior prevalence still affects the posterior after the evidence is observed
- c) The evidence term can be omitted when sensitivity is known
- d) Conditioning makes the prior irrelevant

Answer: _____

Question 5. (5 pts) If a continuous variable has CDF F , which expression gives the probability of an interval $a < X \leq b$?

- a) $F(a) - F(b)$
- b) $F(b) - F(a)$
- c) $f(b) - f(a)$
- d) $F(a)F(b)$

Answer: _____

Question 6. (5 pts) Let $I = \mathbf{1}_A$ be an indicator. Which identity is always true?

- a) $E[I] = P(A)$
- b) $E[I] = P(A)^2$
- c) I must be independent of every variable
- d) $\text{Var}(I) = P(A)$

Answer: _____

Question 7. (5 pts) If $X \sim \mathcal{N}(-2, 1)$ and $Z = -3X + 4$, what is the variance of Z ?

- a) -3
- b) 1
- c) 3
- d) 9

Answer: _____

Question 8. (5 pts) If $U \sim \text{Uniform}(0, 1)$ and $X = F^{-1}(U)$ for a continuous CDF F , what is the purpose of the construction?

- a) It produces a sample with CDF F
- b) It makes every value of X equally likely
- c) It removes the need to define a CDF
- d) It works only for Gaussian distributions

Answer: _____

Part B: Medium Question 9 [20 points]

Question 9. (20 pts) A monitoring model records the joint PMF of status S and alert A :

	$A = 0$	$A = 1$
$S = \text{normal}$	0.45	0.05
$S = \text{fault}$	0.15	0.35

- (a) [4 marks] Give the marginal PMF of A .
- (b) [6 marks] Compute $P(S = \text{fault} \mid A = 1)$ and explain in words what population this conditional probability refers to.
- (c) [6 marks] Test whether S and A are independent using one joint-probability comparison.
- (d) [4 marks] A modeller proposes a coarse sample space with only the outcomes {normal, fault}. Explain one question about alerts that this coarse model cannot answer.

Part C: Medium Question 10 [20 points]

Question 10. (20 pts) Let R have density $f_R(r) = 2(1 - r)$ for $0 \leq r \leq 1$ and 0 otherwise. A service has 12 independent-looking but not necessarily independent requests; request i is rejected with event B_i , with $P(B_i) = 0.08$.

- (a) [6 marks] Derive the CDF $F_R(r)$ on $[0, 1]$ and compute $P(R > 0.5)$.
- (b) [8 marks] Let $M = \sum_{i=1}^{12} \mathbf{1}_{B_i}$. Compute $E[M]$ and explain why the calculation remains valid even if the rejection events are dependent.
- (c) [6 marks] Suppose a two-group measurement model has $Y|G=0 \sim \mathcal{N}(10, 1)$ and $Y|G=1 \sim \mathcal{N}(20, 4)$. Explain why the mixture is not generally determined by its overall mean and variance alone, and give one feature that a single Gaussian can hide.

Part D: Hard Question 11 [40 points]

Question 11. (40 pts) Let X have density $f_X(x) = 3x^2$ for $0 < x < 1$. Define a folded score $Y = \begin{cases} x & 0 \leq x \leq \frac{1}{2} \\ 1-x & \frac{1}{2} < x \leq 1 \end{cases}$ when $X = x$.

- (a) [14 marks] Derive $F_Y(y) = P(Y \leq y)$ by identifying all X -values that map into the event. Give the full piecewise CDF.
- (b) [10 marks] Differentiate to obtain $f_Y(y)$ on the interior of the support and verify that it integrates to one. Explain how the two branches contribute different amounts of mass.
- (c) [8 marks] Since $F_X(x) = x^3$ on $[0, 1]$, describe inverse-transform sampling for X . For $U = 0.125$, compute X and the resulting Y .
- (d) [8 marks] State two checks that would distinguish a correct folded-score implementation from an implementation that accidentally keeps only the $X \leq \frac{1}{2}$ branch. Explain why checks based on finite simulation are not proofs.

End of Validation Set — Check answers with AI and discuss